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RESEARCH ARTICLE

YOLO-Helmet: A Novel Algorithm for Detecting Dense Small Safety Helmets in Construction Scenes

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ABSTRACT Safety helmet wearing is an effective measure for reducing construction safety accidents. However, the current algorithms for detecting helmet-wearing face several challenges, including high missed detection rates and low accuracy in detecting dense small safety helmets. Therefore, this paper proposes a novel algorithm called YOLO-Helmet. Firstly, in order to solve the problem of difficult detection due to the small area of the helmet in the image, a small size detection layer was extended to improve the detection sensitivity of the network to small size targets. Secondly, in order to reduce the influence of occlusion on the accuracy of helmet detection, the C-ELAN module was constructed, and the receptive field is expanded by deformable convolution to provide rich contextual feature information for coordinate attention, so as to improve the accuracy of the network for the discrimination of target position information. Thirdly, CIoU was combined with NWD to reduce the sensitivity of position deviation while retaining the excellent classification ability of CIoU. Finally, in order to facilitate the model deployment, the VoV-DG module based on GSConv was constructed in the neck. The experimental results show that the YOLO-Helmet algorithm achieved an average detection accuracy of 93.1% on the SHWD dataset and is more suitable for the identification of dense small helmets in construction scenes than other mainstream algorithms.

INDEX TERMS Helmet wearing detection, YOLOv7-tiny, coordinate attention, GSConv, NWD.

I. INTRODUCTION

With the rapid progress of urbanization, China's construction industry is booming. Nowadays, the public has paid more and more attention to the safety issues of construction sites. As an important protective equipment on construction sites, safety helmets play a role in effectively reducing the harm to workers' heads caused by falling objects from high places, thus significantly reducing the incidence of accidents on construction sites. However, due to the lack of supervision mechanisms and workers' safety awareness, safety accidents occur frequently [1]. Therefore, the introduction of helmet wearing detection algorithm to monitor workers' wearing of safety helmets in real time has important practical application value.

Early research mainly relied on image processing technology to construct features related to safety helmets for recognition. For example, Liu and Ye [2] used skin color as the main feature to locate human faces and judged whether a safety helmet was worn by comparing the color difference between the human face and the safety helmet. Rubaiyat et al. [3] selected the helmet shape as the feature and applied the Histogram of Oriented Gradients (HOG) to detect helmets. Shrestha et al. [4] used edge detection algorithms to extract contour features of helmets for detection. Wu and Zhao [5] combined hybrid descriptors and hierarchical support vector machines (HSVM) to identify safety helmets. Although these studies have achieved certain results in safety helmet detection, they still have limitations in the ability to extract complex features and generalization ability, which are not very suitable for practical application scenarios.

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In recent years, machine vision technology has witnessed rapid development, leading to the increasing popularity of object detection algorithms based on deep learning among scholars. These algorithms are favored for their simple network structure, fast detection speed, and high accuracy, making them widely applicable in various fields. Deep learning-based object detection algorithms can be broadly categorized into two types: the two-stage detection algorithm represented by the R-CNN series [6], [7],which employs convolutional neural networks to generate candidate regions for screening, classification, and position correction with high precision but limited real-time monitoring capability; and the one-stage detection algorithm represented by SSD [8], YOLO series [9], [10], [11], and RetinaNet series [12], which directly perform regression and classification resulting in faster detection speed. In the field of safety helmet detection, scholars have conducted extensive research and achieved remarkable results. For example, Zhang et al. [13] addressed the challenges of detection in complex scenes by introducing the SPP structure and the CBAM attention mechanism, thereby building the SCM-YOLO framework. Compared with the original model, this framework improved the detection accuracy by 4.76%, significantly enhancing the safety helmet detection capability in complex environments. Chen et al. [14] focused on the convenience of model deployment in practical applications, especially on edge devices. They proposed PP-LCNet as the backbone network, based on and optimized the Depthwise Separable Convolution, which was originally proposed by MobileNetV1. This innovation not only significantly reduced the size of the model (about 83%), but also improved its operating efficiency and applicability on edge devices. Han et al. [15] targeted the characteristics of the construction industry, especially the potential problem of low resolution in images, and designed a novel Super-Resolution (SR) reconstruction module. This module, combined with the CSP module, effectively reduced the risks of information loss and gradient confusion, making the model more accurate and reliable in safety helmet detection. Song and Wang [16] proposed the RBFPDet algorithm, which utilizes strong semantic feature points for safety helmet detection. By capturing and utilizing key feature points in images, the RBFPDet algorithm improved the accuracy of safety helmet detection while maintaining detection speed. Despite significant progress made in these studies, there are still many challenges in practical applications, such as occlusion in crowds, few helmet pixels in images, and potential feature information loss during network training. These challenges become particularly significant when dealing with dense small targets.

To address these issues, we propose a new detection algorithm, YOLO-Helmet, based on the YOLOv7-tiny network model. The following are the main innovations and contributions of this paper:

1) In order to improve the detection effect of the network on the occluded safety helmet, we replace the 3×3 ordinary convolution in the ELAN module with the flexible deformable convolution. In addition, we introduce the coordinate attention mechanism after this convolutional layer to build the C-ELAN module. This design aims to enlarge the receptive field and capture more context information, so that the occluded target can be located more accurately.
2) To reduce the cost of network deployment, this paper uses GSConv to build the VOV-DG module in the neck. The global sparsity of GSConv is used to reduce the computational burden of the network while maintaining the accuracy.
3) In order to alleviate the performance decline caused by the drastic fluctuations of CIoU loss during training, we have combined CIoU loss with NWD loss. This integration aims to stabilize the training process while leveraging the advantages of CIoU loss in measuring bounding box overlap and shape differences.
4) Addressing the issue of distant targets occupying small areas in images, we have added a fine-grained detection layer to increase the network's sensitivity to small-sized targets, thereby overall improving detection performance.

II. METHODOLOGY AND IMPROVEMENT

A. YOLOv7-TINY

YOLOv7-tiny, as a lightweight version of YOLOv7, possesses outstanding real-time performance. The structure of this network is shown in Figure 1.

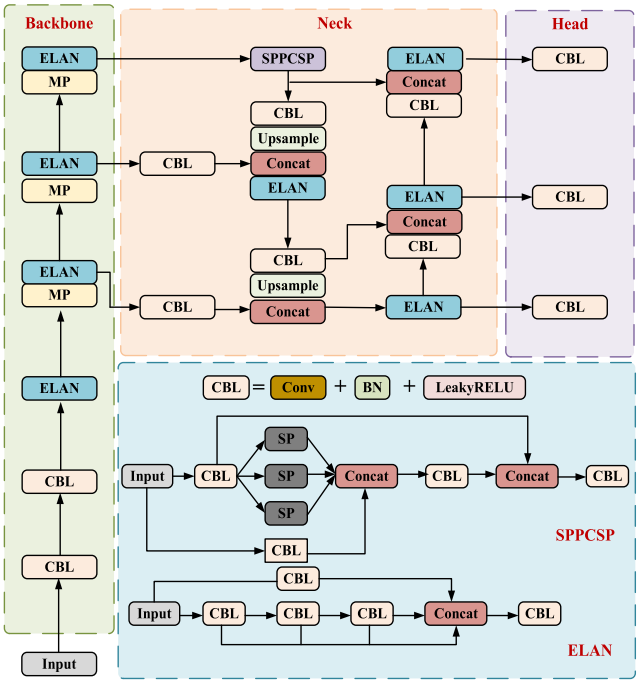


FIGURE 1. The network architecture of the YOLO-tiny.

First, data augmentation is applied to the image as it is input. Then, the processed image is transmitted to the backbone network constructed by CBL, ELAN and MP,

and feature information is extracted from it. Next, the PANet [17] architecture through the neck fully fuses the extracted multi-scale information. The final output is the class information, confidence score, and regression information for the target object.

B. C-ELAN

In the field of computer vision, attention mechanisms play a crucial role in optimizing network performance by assigning appropriate weights to each position of the feature map, guiding the network model to focus on key features and significantly enhancing the model's expressive power and detection accuracy. Numerous studies have demonstrated the significant impact of integrating attention modules in improving neural network performance in dense small object detection, especially in the presence of occlusions. For instance, Liu et al. [18] introduced the CBAM attention mechanism into the backbone network to enhance the detection accuracy of small objects in aerial images. CBAM allocates reasonable weights to spatial and channel information in the feature map, effectively suppressing background interference, leading to a 2.3% increase in detection accuracy. In another study, Di et al. [19] constructed a lightweight TP-YOLO network for detecting tiny pests by incorporating two attention-based components. Experimental results showed that this network outperformed state-of-the-art methods in both detection accuracy and inference time. However, considering that occluded objects may be challenging to accurately locate based solely on existing features, especially in applications like safety helmet detection, this paper introduces the high-performance Coordinate Attention (CA) [20] mechanism into the network model to address this issue. The CA mechanism enhances the correlation between the position information of safety helmets and spatial information, helping the network to more accurately locate occluded safety helmets. The structure of the coordinate attention module is illustrated in Figure 2. Firstly, it performs encoding operations on input feature channels in both horizontal and vertical directions to capture critical information in these two directions. Subsequently, the module merges the information from these two directions, generating a feature map that integrates both horizontal and vertical features. Then, utilizing methods such as split and convolution, the module can focus on the precise position of the target in both horizontal and vertical directions. Ultimately, the module outputs the positional information of the target of interest (such as a safety helmet), which is directly utilized for subsequent detection and recognition tasks.

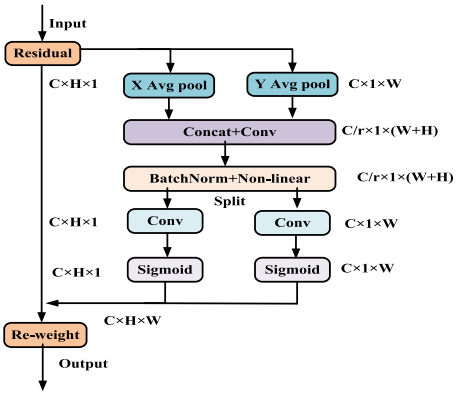


FIGURE 2. The structure of the CA.

Deformable Convolution (DC) allows individual elements of the convolutional kernel to dynamically adjust based on input features, aiding in capturing objects of different shapes and sizes in images, expanding the receptive field, and enhancing feature extraction capability. Therefore, by replacing the original two 3×3 large convolutions in the ELAN module with deformable convolutions, spatial features and contextual information in images can be more effectively extracted. Subsequently, introducing the Coordinate Attention (CA) mechanism enables precise focus on the position and channel of features, strengthening and extracting key task features. This approach significantly boosts model performance without significantly increasing computational costs. The resulting module is named C-ELAN, and its structure is illustrated in Figure 3. The detection effect comparison before and after introducing this module is shown in Figure 4. As can be seen from the figure, after introducing the C-ELAN module, the detection accuracy of safety helmets has been significantly improved, and the missed detection rate has also been reduced accordingly.

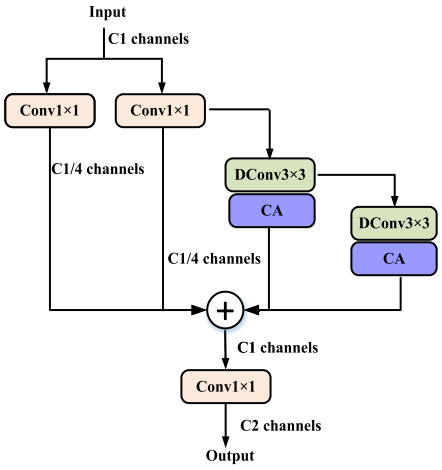


FIGURE 3. The structure of the C-ELAN.

C. VoV-DG

When deploying real-time detection tasks on mobile devices and embedded systems, the real-time performance becomes a critical consideration. To meet this demand, researchers have introduced a series of lightweight network structures such as Xception [21], MobileNet [22], and ShuffleNet [23]. These network architectures leverage Depthwise Separable Convolution (DSC) technology to decompose standard convolutions into depthwise convolutions and pointwise convolutions, significantly reducing the model complexity and



FIGURE 4. Comparison of detection effect diagrams before and after introducing C-ELAN.

computational cost. However, the separation of channels during DSC restricts the interaction of information between channels to some extent, thereby affecting detection accuracy. For instance, Mei et al. [24] replaced the backbone network of YOLOv7 with MobileNetv3 to reduce computational load. Although this modification led to a reduction in the parameter count by 30.37×10^6, the average precision of the model decreased by 10.6%. Similarly, Wu et al. [25] attempted to substitute the C3 module in the feature extraction network of YOLOv5 with ShuffleNetV2's basic residual structure and replaced the CBS module with a Depthwise Separable Convolution module. While this improvement reduced the parameter count and floating-point operations of the lightweight network YOLOv5-Light by 61.6% and 75%, respectively, the detection accuracy decreased by approximately 3.8 percentage points. To avoid sacrificing accuracy for speed, we introduce an enhanced convolutional structure called Ghost-Shuffle Conv (GSConv) [26]. GSConv achieves effective fusion of channel information by convolving sparse positions of the feature map. The design of the GSConv module is illustrated in Figure 5. It first generates a set of new feature maps (referred to as "ghost feature maps") through a 1×1 convolution. Then, it utilizes DSC to process the input feature maps and concatenates the processed features with the ghost feature maps. Finally, a channel shuffle operation rearranges the channels of the concatenated feature maps, enabling effective information exchange between channels and subsequently improving the detection accuracy of the model.

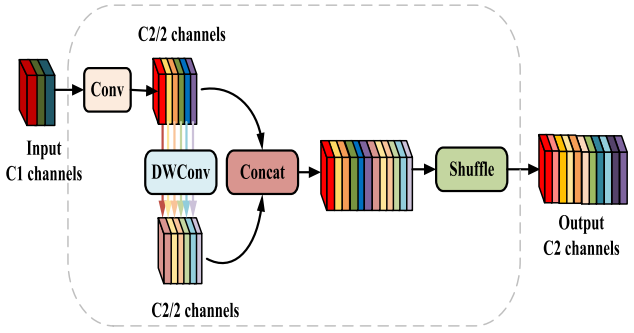


FIGURE 5. The structure of the GSConv.

The ELAN module, which is composed of multiple stacked standard convolutions, introduces complexity and computational burden to the network structure. However, replacing all convolutions with GSConv would increase the network depth and consequently impact detection speed. To strike a balance between network accuracy and complexity, we propose introducing GSConv in the neck region and designing the VoV-DG module as a substitute for the ELAN module in fusion networks. The structure of the VoV-DG module, as shown in Figure 6, consists of two parallel branches. The first branch utilizes regular convolutions to process the feature maps and maintain basic feature extraction capabilities. The second branch employs a DGBottleneck module that incorporates GSConv, leveraging the lightweight nature of GSConv to reduce the computational burden of the model. Finally, the feature information obtained from both branches is concatenated and further processed by convolutions to obtain the final output of the module.

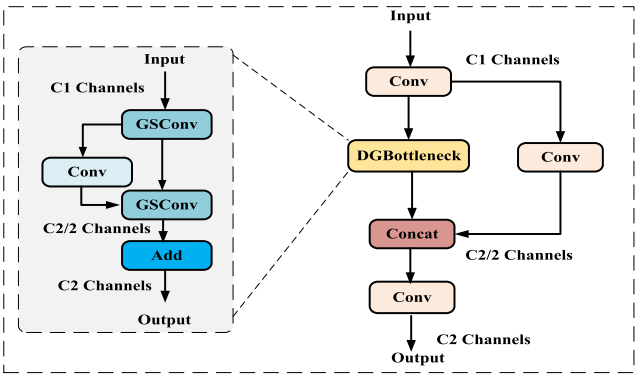


FIGURE 6. The structure of the VoV-DG.

D. NWD LOSS

In YOLOv7-tiny, to more accurately quantify the similarity between predicted bounding boxes and ground truth bounding boxes, the CIoU loss function is traditionally used to calculate the bounding box loss. However, due to the discrete nature of the CIoU loss function in measuring the similarity of bounding boxes, even minor positional deviations can lead to significant changes in the CIoU metric value. This discreteness may result in drastic fluctuations in the loss value during training, thereby affecting the detection performance of the model. To overcome this limitation of the CIoU loss function, we introduce the NWD loss function [27]. The NWD loss function can handle the relative size and positional relationship between bounding boxes more smoothly, providing a more refined quantification of the similarity between target bounding boxes and predicted bounding boxes. In cases of slight positional deviations, the NWD loss can maintain a more stable loss value, reducing the drastic fluctuations in the loss value caused by the discreteness of CIoU. Therefore, to reduce the sensitivity to positional deviations and improve detection performance, we propose a method that combines CIoU loss and NWD loss to calculate the bounding box loss. This approach retains the advantages of CIoU loss in measuring the overlap and shape differences of bounding boxes while compensating for its discreteness through the introduction of NWD loss.

The NWD loss is a distance metric that measures the difference between two probability distributions. Firstly, the predicted boxes and ground truth boxes are modeled by a two-dimensional Gaussian distribution. Then, Wasserstein distance is used to calculate the similarity between the predicted and ground truth boxes, resulting in NWD loss. Specifically, assuming that the center coordinates of box $R = (c_x, c_y, w, h)$ are (c_x, c_y), width is w, and height is h, we first model box R and obtain its Gaussian distribution $N(\mu, \sum)$ through equation (1). Next, we compute the distribution distance between bounding box $R_1 = (c_{x1}, c_{y1}, w_1, h_1)$ and $R_2 = (c_{x2}, c_{y2}, w_2, h_2)$ using the Wasserstein distance formula (2). Lastly, we obtain the NWD loss value through equation (3).

$$\mu = \begin{bmatrix} c_x \\ c_y \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \frac{w^2}{4} & 0 \\ 0 & \frac{h^2}{4} \end{bmatrix} \tag{1}$$

$$w_2^2 = \left\| \left(\left[c_{x1}, c_{y1}, \frac{w_1}{2}, \frac{h_1}{2} \right]^T, \left[c_{x2}, c_{y2}, \frac{w_2}{2}, \frac{h_2}{2} \right]^T \right)^T \right\|_2^2 \tag{2}$$

$$L_{NWD} = \exp\left(-\frac{\sqrt{w_2^2}}{C} \right) \tag{3}$$

The CIoU loss calculates its value based on overlap area of bounding boxes, center point distance, and aspect ratio as shown in equation (4).

$$L_{CIoU} = IoU - \left(\frac{\rho^2\left(b, b^{gt}\right)}{c^2} + \alpha v \right) \tag{4}$$

where $\rho^2\left(b, b^{gt}\right)/c^2$ represents a penalty term; b denotes predicted box center coordinates; b^{gt} represents ground truth box center coordinates; $\rho^2(\bullet)$ indicates Euclidean distance; c represents diagonal distance between b and b^{gt} smallest enclosing rectangle; α is a positive balancing parameter; v signifies consistency of aspect ratio between gt and predicted box.

The overall localization loss calculation formula is given by equation (5).

$$L_{box} = \beta L_{NWD} + (1 - \beta) L_{CIoU} \tag{5}$$

where L_{CIoU} represents CIoU loss while L_{NWD} stands for NWD loss. β denotes weight coefficient for NWD loss. After experimental verification, the best effect is achieved when the value of β is 0.7.

E. ADD A SMALL TARGET DETECTION LAYER

With an input size of 640×640, the scatter plot distribution of all target sizes in the dataset is shown in Figure 7. This dataset primarily focuses on medium and small-scale safety helmets as the target of detection. Given that the minimum detectable size of the YOLOv7-tiny network is 16×16, when the size of the target to be detected is smaller than this threshold, the model may have difficulty identifying it, resulting in missed detections. However, in actual construction scenes, workers are distributed in various corners, including those regions far away from the camera, which leads to relatively small target sizes in the image. To enhance the network's ability to recognize small targets, this paper proposes extending a small detection layer in the YOLOv7-tiny network. This improvement aims to increase the network's sensitivity to small-sized targets, thereby improving overall detection performance, especially for detecting small safety helmets at a distance in actual construction scenes. The final obtained model YOLO-Helmet is shown in Figure 8.

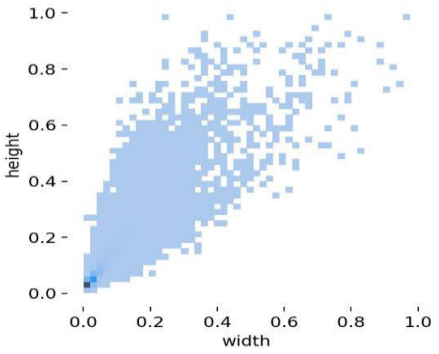


FIGURE 7. Scatter plot of target size distribution.

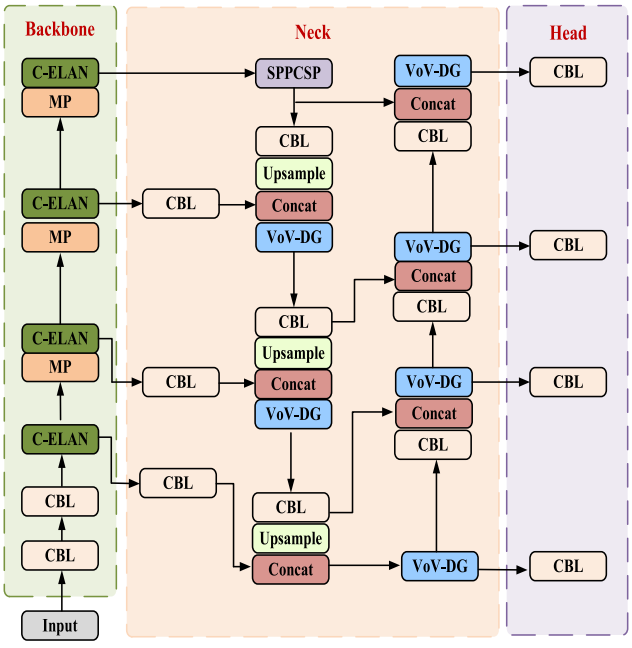


FIGURE 8. The network architecture of the YOLO-Helmet.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. DATASET COLLECTION AND PROCESSING

The dataset used in the experiment is from the SHWD dataset (safety helmet wearing-dataset) publicly available on the

Internet, which contains 7581 images and has been labeled with two categories: helmet and no-helmet. Among them, helmet represents the personnel wearing a safety helmet, and no-helmet represents the personnel not wearing a safety helmet. Since the annotation format was VOC format, which was not suitable for model training, it was converted to YOLO format using Python script and randomly divided into 5957 training set, 812 validation set and 812 test set according to 8:1:1 ratio.

B. EXPERIMENTAL ENVIRONMENT

Both the training and testing of this experiment were carried out using the Windows 10 operating system. The specific software and hardware environments are listed in Table 1.

TABLE 1. Experimental environmental configuration.

Configuration Name	Version Parameters
Operating system	Windows10
Graphics (GPU)	GeForce RTX2070SUPER
Processor (CPU)	Intel® Core™ i9-9900
Framework	Pytorch(1.11.0)
GPU acceleration environment	CUDA11.6+cudnn8.4.1
Processor (CPU)	Intel® Core™ i9-9900

C. EXPERIMENTAL HYPERPARATION AND EVALUATION CRITERIA

To ensure the effectiveness of the experimental data, the same hyperparameters were used for training and testing different algorithm models. The parameter settings are as follows: an initial learning rate of 0.01, weight decay set at 0.00002, a batch size of 16, and a momentum factor of 0.937 to prevent model training from getting stuck in local optima; additionally, the number of epochs is set to be 200. Figure 9 shows the comparison curve of loss values between YOLOv7-tiny and YOLO-Helmet. The loss value comparison curve between YOLOv7-tiny and YOLO-Helmet training is depicted in Figure 10. It can be observed that, in contrast to YOLOv7-tiny, YOLO-Helmet exhibits convergence at a lower loss value, indicating effective acquisition of target features during the training process and attainment of favorable fitting outcomes.

To objectively evaluate the model's detection performance, we employ precision (P), recall (R), and mean average precision (mAP) as evaluation indicators, while set intersection over union threshold of 0.5 for calculating mAP. The formulas for these metrics are as follows:

$$P = \frac{TP}{TP + FP} \tag{6}$$

$$R = \frac{TP}{TP + FN} \tag{7}$$

$$mAP = \frac{\sum_{i=1}^{k} AP_i}{k} \tag{8}$$

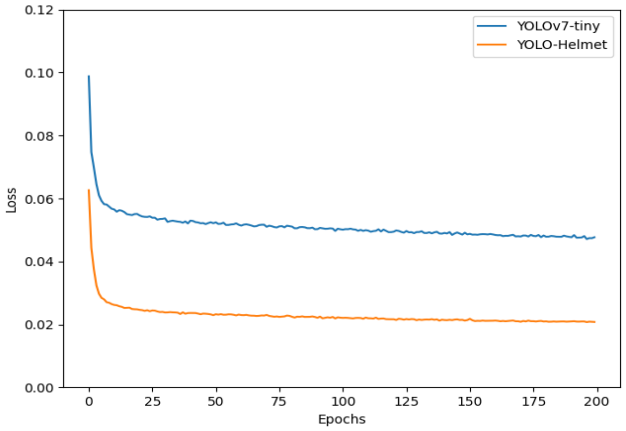


FIGURE 9. Loss values contrast curve.

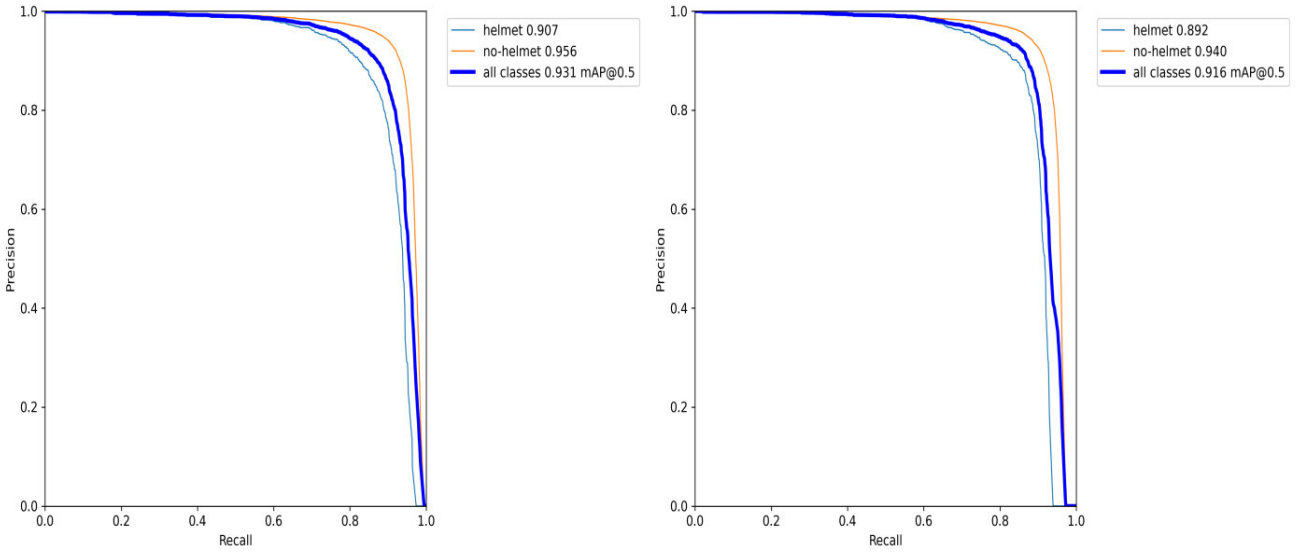
where TP represents true positive cases, FP represents false positive cases, FN represents false negative cases, k represents the number of classes, and AP_i represents the area under the PR curve for class i. The PR curves of different categories for YOLOv7-tiny algorithm and YOLO-Helmet algorithm are shown in Figure 9. Obviously, YOLO-Helmet algorithm consistently achieves higher mAP values across all categories, indicating that YOLO-Helmet is more robust than YOLOv7-tiny.

D. COMPARISON EXPERIMENT OF IMPROVED MODULE PERFORMANCE

During the construction of the C-ELAN module, we employed the CA. To verify that CA has stronger detection performance than other attention mechanisms in the task of safety helmet detection, this paper conducted comparative experiments, comparing CA with classic SE, SA, EMA, and CBAM. To ensure the reliability and consistency of the experimental results, all experiments were conducted on the same dataset and under the same experimental environment. The experimental results are shown in Table 2. Based on the data in the Table 2, it is evident that the CA outperforms other attention mechanisms in the safety helmet detection task. Specifically, it achieves the highest values in key metrics such as precision, recall, and mAP, with 91.8%, 88.1%, and 92.3% respectively. This indicates that the CA possesses high accuracy and comprehensiveness in safety helmet detection, demonstrating excellent positioning and recognition capabilities. Additionally, there are minor differences in the number

TABLE 2. Comparison of attention performance.

Attention	P/%	R/%	mAP/%	Params/M
SE	91.6	87.9	92.2	6054206
SA	90.7	87.3	91.6	6018030
EMA	91.8	86.5	91.6	6066142
CBAM	90.3	87.6	91.4	6055238
CA	91.8	88.1	92.3	6048630



(a) PR curve of the YOLOv7-tiny algorithm

(b) PR curve of the YOLO-Helmet algorithm

FIGURE 10. Loss values contrast curve. PR curve.

of model parameters among the various attention mechanisms. In summary, the CA exhibits the best performance in the safety helmet detection task.

In order to verify the superiority of the location of CA attention mechanism in C-ELAN module, this paper carries out comparative experiments on different locations of CA. It is divided into two groups, the first group is behind the whole ELAN module, and the second group is behind the 3×3 convolution inside ELAN. The experimental results are shown in Table 3. According to the data in Table 3, the number of network parameters of CA attention mechanism is similar when it is placed in the two positions. However, when CA is placed behind the 3×3 convolution inside ELAN, the accuracy, recall and precision are higher than those behind ELAN, which are 0.2%, 1.2% and 0.6% higher respectively. This indicates that placing CA after the inner convolution of ELAN enables more efficient use of local features and better integration of these features in subsequent processing, thus improving the overall performance of the model.

TABLE 3. Performance comparison of CA in different positions.

Group	P/%	R/%	mAP/%	Params/M
1	91.6	86.9	91.7	6035548
2	91.8	88.1	92.3	6048630

During the construction of the lightweight module VOV-DG, we decided not to employ depthwise separable convolution but instead chose the more performant GSConv as the core component. To validate the correctness of this choice, we conducted comparative performance experiments by utilizing GSConv, DSC, and another lightweight convolution method, spatial separable convolution (SSC), as the core modules. To ensure the reliability and consistency of the experimental results, all experiments were conducted on the same dataset and under identical experimental conditions. The experimental results are shown in Table 4. Based on the data in the Table 4, when using depth separable convolution, there is a significant reduction in the number of parameters, but other metrics such as recall and precision show a decrease as well. This indicates that depth separable convolution decomposes the traditional convolution process into two separate steps: depthwise convolution and pointwise convolution. It successfully achieves channel isolation, thereby reducing the complexity of the model. However, this also results in a lack of effective information exchange between channels, leading to a noticeable decrease in the model's detection performance. In contrast, with spatial separable convolution, although precision slightly decreases compared to the baseline model, recall remains similar and mAP shows a slight improvement. The parameter count also decreases, although not to the extent seen with depth separable convolution. GSConv maintains baseline performance levels in terms of precision and recall, but shows a significant improvement in mAP while also reducing the number of parameters.

TABLE 4. Comparison of lightweight convolution performance.

Module	P/%	R/%	mAP/%	Params/M
YOLOv7-tiny	91.1	87.1	91.6	6017694
+DSConv	91.2	86.7	91.3	5706414
+SSConv	90.7	86.9	91.7	5798822
+GSConv	91.2	87.1	91.9	5747622

This highlights that GSConv achieves model lightweighting while maintaining performance, avoiding the issues seen with depth separable convolution in sacrificing performance to enhance model deployability. In conclusion, selecting GSConv as the core component of VOV-DG is a wise decision.

In order to verify whether replacing GSConv only in the neck has better performance than replacing the whole network convolution, this paper conducts a comparison test. The experiments were divided into two groups: the first group replaced the whole network convolutions with GSConv, and the second group replaced only the neck convolutions with GSConv. The experimental results are detailed in Table 5. According to the data in Table 5, the number of parameters in the first group is 291319 more than that in the second group, indicating that when all convolutions of the network are replaced by GSConv of the neck, the model depth will be increased and the detection speed will be affected. However, the accuracy and recall are higher when replacing only at the neck, which indicates that this move can reduce the possibility of missed detection and improve the detection accuracy. In summary, the performance is better when only the neck convolution is replaced.

TABLE 5. Performance comparison of GSConv in different positions.

Group	P/%	R/%	mAP/%	Params/M
1	91.1	86.7	92.0	6038941
2	91.2	87.1	91.9	5747622

To verify the superiority of combining CIoU loss and NWD loss compared to using them individually, this study conducted a comparative experiment, and the experimental results are shown in Table 6. Observing the data in the Table 6, it is evident that when these two losses are combined, the target recognition accuracy and precision are higher compared to when used individually. Therefore, the conclusion can be drawn that combining CIoU and NWD losses yields superior results compared to using them separately.Meanwhile, it can be seen from Figure 11 that the detection effect is the best when is 0.7.

TABLE 6. Comparison of loss function performance.

Loss	P/%	R/%	mAP/%	Params/M
CIoU	91.1	87.1	91.6	6017694
NWD	91.4	85.0	90.8	6010302
CIoU+NWD	91.6	86.7	92.0	6010302

E. ABLATION EXPERIMENT

A series of ablation experiments were conducted in this paper to validate the effectiveness of the YOLO-Helmet algorithm. To provide a more intuitive evaluation of the improvement effect of each module, the experiments were divided into six groups. The performance indicators for each group of experiments are presented in Table 7.

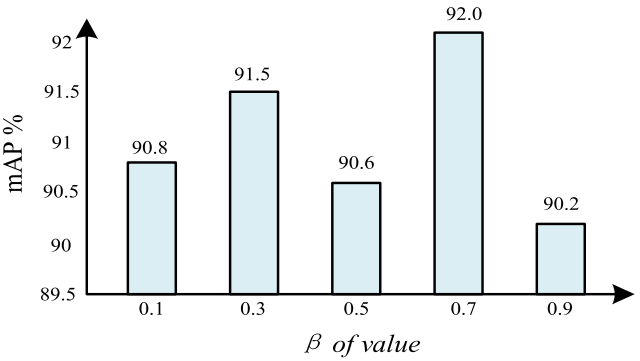


FIGURE 11. The effect of x on y.

TABLE 7. Results of ablation experiments.

Group	C-ELAN	VoV-DG	NWD	Small	mAP/%	Params/M
1					91.6	6017694
2	√				91.9	6270870
3		√			91.9	5747622
4			√		92.0	6010302
5				√	92.5	6518238
6	√	√	√	√	93.1	6575812

As observed in Table 7, with the implementation of the optimization strategy, the detection performance of YOLOv7-tiny is gradually improved. The incorporation of the C-ELAN module slightly increases the network parameters, yet effectively improves the recognition rate of safety helmets, indicating successful localization of target position information through embedded attention mechanism. Replacing the ELAN module with the VoV-DG module reduces network parameters by 270072M and increases mAP by 0.3%, demonstrating that this improvement does not have the drawback of depthwise separable convolution. By combining original bounding box loss and NWD loss, recognition accuracy surpasses 92.0%. Furthermore, incorporating a detection layer leads to an additional accuracy improvement of nearly 1%, resulting in an overall recognition accuracy of 92.5%. After comprehensively applying all improved methods to the YOLOv7-tiny model, the mAP value surpasses 93%, reaching 93.1%. This is a significant increase compared to YOLOv7-tiny's mAP value and highlights YOLO-Helmet's potential for reducing false detections and missed detections while demonstrating higher reliability in object detection.

To further verify and analyze the effect of the YOLO-Helmet model in detecting densely occluded targets, the Gradient-Weighted Class Activation Mapping (Grad-CAM) [28] is used to generate heat maps between the proposed model and the original model. Grad-CAM is a gradient-based method that visualizes the features obtained by deep neural networks through computing the weight of each feature map, deriving the global average of gradients, and executing backward propagation to obtain gradient values.

Such an approach facilitates the analysis of the network's focal regions concerning a particular class, allowing for a retrospective assessment of whether the network has accurately acquired specific features or information. Specifically, in the heatmap, deeper shades of red indicates that the network pays more attention to this part of the image, while deeper shades of blue means that the network considers this information to be redundant.

Figure 12 presents the experimental results. It can be inferred that YOLOv7-tiny demonstrates limited attention towards dense objects and struggles to effectively focus on the target area. Furthermore, there are instances of false detections, such as mistakenly identifying rocks as the intended targets for detection. In contrast, the model proposed in this paper demonstrates effective noise suppression in the background and accurately focuses on the central point of the target. Therefore, it can be seen that YOLO-Helmet performs better in detecting dense safety helmets.



(a) Original image (b) YOLOv7-tiny heatmap (c) YOLO-Helmet heatmap

FIGURE 12. Heatmap comparison.

F. COMPARATIVE EXPERIMENT

To authenticate the performance of the YOLO-Helmet algorithm, this paper makes an experimental comparison with the two-stage mainstream detection algorithm Faster-RCNN and the one-stage mainstream detection algorithms SSD, YOLOv4, YOLOv5s, YOLOv7-tiny. The experimental results are presented in Table 6, and a histogram is plotted in Figure 13 to facilitate the comparison of the results.

From Figure 13, it is evident that YOLOv5s, YOLOv7-tiny, and YOLO-Helmet exhibit superior performance compared to other algorithms. Furthermore, upon analyzing Table 8 in detail, it becomes apparent that both YOLOv5s (91.8%) and YOLO-Helmet (91.5%) demonstrate high accuracy in object classification, showcasing their exceptional ability to precisely classify objects during detection tasks. In terms of recall rate, YOLOv7-tiny (87.1%) and YOLO-Helmet (86.9%) perform admirably well by effectively capturing existing objects with efficiency. Regarding mAP, the standout

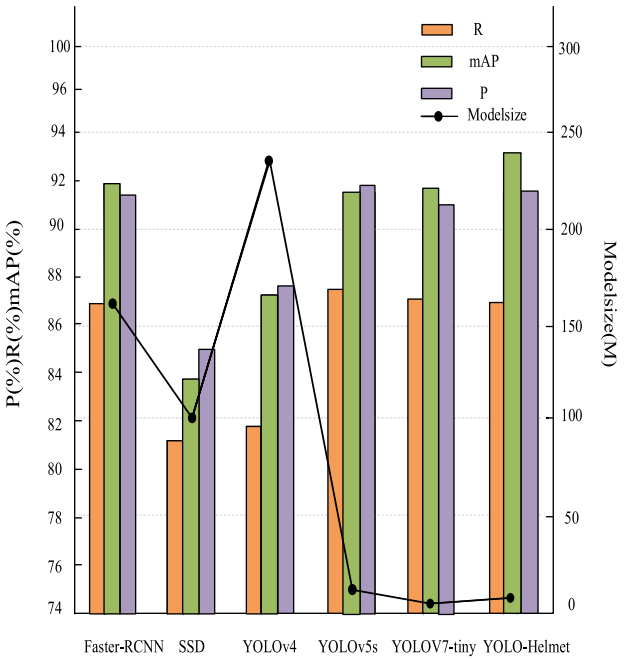


FIGURE 13. Compares with current mainstream detection algorithms.

TABLE 8. Compares with current mainstream detection algorithms.

Algorithms.	P/%	R/%	mAP/%	Modelsize/M
Faster-RCNN	91.4	86.9	91.9	160
SSD	85.0	81.2	83.7	101.2
YOLOv4	87.6	81.8	87.2	235
YOLOv5s	91.8	87.5	91.5	14.4
YOLOv7-tiny	91.1	87.1	91.6	6.2
YOLO-Helmet	91.5	86.9	93.1	3.6

performer is YOLO-Helmet at 93.1%, closely followed by YOLOv5s at 91.5%. This emphasizes their remarkable accuracy in recognizing various object categories across different scenarios. Additionally, although YOLO-Helmet is 2.1M larger than YOLOv7-tiny, it is still relatively efficient in storage and deployment.

In summary: Compared to other detection algorithms, YOLO-Helmet exhibits superior detection performance. Furthermore, Figure 14 showcases the exceptional ability of YOLO-Helmet in detecting dense safety helmets, surpassing both YOLOv5s and YOLOv7-tiny. In Figure 14(a), YOLO-Helmet successfully detects the obscured red helmet nearby, in contrast to the inability of YOLOv5s to accurately recognize it. By observing Figure 14(b), it can be seen that YOLO-Helmet outperforms other algorithms in terms of helmet recognition accuracy, boasting a recognition rate exceeding 90%. Moving on to Figure 14(c), where dense helmets at medium and long distance are analyzed, YOLOv7-tiny fails to identify the helmet-less individual at the top right corner. In contrast, both YOLO-Helmet and YOLOv5s successfully pinpoint the target with higher overall accuracy



(a) close-range dense safety helmets

(b) medium-distance dense safety helmets

(c) medium and long distance dense safety helmets

(d) long distance dense safety helmets

FIGURE 14. Comparison of detection results in the safety helmet test dataset. From left to right, the images show the original image, YOLOv5s detection result image, YOLOv7-tiny detection result image, and YOLO-Helmet detection result image; from top to bottom, they display images of close-range dense safety helmets, images of medium-distance dense safety helmets, images of medium and long distance dense safety helmets, and images of long distance dense safety helmets awaiting testing.

exhibited by YOLO-Helmet. Finally, in the context of dense helmets at long distances depicted in Figure 14(d), both YOLOv5s and YOLOv7-tiny exhibit noticeable areas of missed detection while minimal regions of missed detection are observed with YOLO-Helmet. In summary, when it comes to detecting dense helmets, YOLO-Helmet demonstrates better applicability and effectively solves the problem of missed detections caused by dense occlusion.

IV. CONCLUSION

In response to the issues of missing dense targets and low accuracy in existing helmet-wearing detection algorithms, a new detection algorithm named YOLO-Helmet has been proposed. Firstly, a small-scale detection layer is added to the YOLOv7-tiny network to enhance the network's sensitivity towards small-sized targets. Secondly, the C-ELAN module is developed to enhance the identification ability of the network for occluded targets. Additionally, by combining the NWD loss function, the training process is stabilized while retaining the advantages of measuring bounding box overlap and shape differences. Lastly, for ease of model deployment, the VoV-DG module is designed for the neck region. Experimental results demonstrate that the YOLO-Helmet algorithm achieves a 1.5% increase in average accuracy compared to the original algorithm. Furthermore, when compared with other algorithms, the YOLO-Helmet algorithm exhibits superior detection performance, effectively reducing the missed detection rate of dense small helmets. However, as seen in Figure 13, YOLO-Helmet still faces some missed detection issues. Therefore, efforts will continue to address this problem and provide a more optimized surveillance solution for construction sites.

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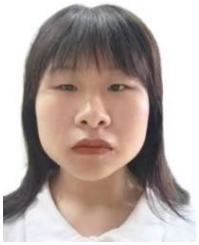
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